

STEFANO BUDI

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EDUCATION

BINUS University

B.S. in Computer Science, Data Science Specialization Track

GPA: 3.97 / 4.00

Jakarta, Indonesia

2024 – 2028 (Expected)

Independent Study: MIT 15.401 Finance Theory I, Harvard STAT110 Probability, Stanford CS229 Machine Learning

PUBLICATIONS

“Dynamic Classifier Selection using Local Class Accuracy on Overlap Cleaned Heterogeneous Pool of Experts”

2nd Author | Published

Sept 2026

- Co-developed a dynamic classifier selection framework combining KNN-based overlap cleaning with local-class-accuracy-based expert selection across a heterogeneous pool of specialists (Naive Bayes, XGBoost, SVM), cutting log loss by 70.1% (0.898 → 0.269) versus a probability-harvesting baseline while achieving 91.62% accuracy and a macro F1-score of 0.8638 on the UCI Student Dropout dataset (n=4,424).

EXPERIENCE

Laboratory Assistant, BINUS University

Feb 2025 – Feb 2026

- Selected as 1 of 2 assistants from ~200 applicants (top <1%) through a 4-month, 3-stage process, including 1 month of full-time (7am–5pm) on-site bootcamp training, project creation, and presentation.
- Taught 5 classes weekly (~150 students) across Computer Vision and Algorithm & Programming (C), qualified in Deep Learning and Data Mining via BINUS's SLC qualifications; earned a 5.3/6 performance KPI.

PROJECTS

Speech2Market — End-to-End NLP-Based Market Impact Predictor for FED Speeches

Aug 2026

[Dashboard](#) | [Live App](#) | [GitHub](#)

- Engineered a two-stage NLP + time-series pipeline (FinBERT embeddings + lagged features, HistGradientBoosting classifier for sign (+/-) + regressor for magnitude) forecasting SPX, Gold, VIX, and TNX returns across 3/7/30-day horizons following FED speeches; validated features via paired bootstrap significance testing (n=2,000).
- Built an automated MLOps ETL & Vector pipeline using daily GitHub Actions to scrape FED transcripts, vectorize text via ONNX FinBERT into Supabase, and sync macroeconomic/price indicators while handling API rate limits.
- Orchestrated monthly continuous retraining with automated quality gating (variance, boundary, and sign-inversion checks) and performance logging (RMSE, F1, AUC) to deploy models seamlessly within a 1GB memory constraint.

FindIT — Face Anti-Spoofing Detection

Apr 2026

Computer vision competition — camera-based identity fraud detection (DAC Find IT! 2026)

- Built a 5-fold DINOv2-based face anti-spoofing classifier for 6 classes with frozen-backbone head training followed by full fine-tuning with different learning rates; used sample weighting and selective MixUp to solve class imbalance.
- Achieved 0.9742 macro F1, placing 11th of 340 teams in the preliminary round (top 4%) — 0.005 behind 1st place, narrowly missing the top-10 cutoff for finals.

Laptop Price Predictor — End-to-End Ensemble ML Deployment

May 2026

[Live App](#) | [GitHub](#)

- Built a stacking ensemble (XGBoost, LightGBM, CatBoost → RidgeCV meta-learner) tuned via Optuna Bayesian optimization, achieving 6.67% MAPE and outperforming all base models in accuracy and cross-validation stability.
- Shipped a full pipeline from training with asynchronous web-scraped dataset (BeautifulSoup + Playwright, 583 listings) to a deployed Streamlit app (P90 latency ~65–75ms), validated via a 5-user study with 100% “useful” rating.

Makan Bergizi Gratis (MBG) Investment Priority Mapping — Graph-Based Clustering Research

Jul 2025

[Dashboard](#) | [GitHub](#)

- Applied spectral (graph-based) clustering with a Laplacian structure to map investment priority across 38 Indonesian provinces, engineering 15+ derived features from raw government data (BPS, Badan Pangan Nasional), filtered via RFECV, validated via the Calinski-Harabasz index (3.94 vs. a 1.01 random-baseline benchmark).
- Selected by BINUS's Computer Science faculty as product presentation material in Pameran Produk Inovasi (PPI) by Badan Riset dan Inovasi Daerah (BRIDA), presenting to government stakeholders and civil servants.

TECHNICAL SKILLS

Languages: Python, SQL (fluent); C (proficient); Java (basic); RapidMiner, R (familiar)

Machine Learning: Gradient Boosting, Spectral Clustering, Temporal Fusion Transformers, Naïve Bayes, Decision Trees, SVM, PCA/LDA, Optuna

Transfer Learning: DINOv2, ConvNeXt, FinBERT, ONNX Runtime, LLM Application Integration

Data Engineering & Deployment: Pandas, BeautifulSoup, Playwright, GitHub Actions, Streamlit, Git/GitHub, AWS